

Yan Lukashevich

Full-stack Developer — .NET · React · Azure · Python

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PROFESSIONAL PROFILE

Single-handedly designed, built, and shipped to production a commercial self-service printing system (.NET 8, React 19 / TypeScript, Microsoft Azure) — from concept and code, through payments, cloud, and CI/CD, to an institutional client. A broad engineering background: Linux and networking (HPC cluster, Cisco CCNA certificate), two years of scientific-software development and data analysis in Python (NCU), and embedded / IoT (Raspberry Pi). I deliver projects to the finish regardless of whether the barrier is technical, business, or formal — from zero to production in 3 months, a startup-competition win (Copernicus Startup Stars 2026), and a passed public-institution security audit.

TECH STACK

- **Backend:** C#, ASP.NET Core 8, Entity Framework Core, REST API, WebSocket · Node.js, NestJS · Python (FastAPI)
- **Frontend:** React 19, TypeScript, JavaScript, Vite, Tailwind CSS, i18n, accessibility (a11y)
- **Cloud / DevOps:** Microsoft Azure (App Service, SQL Database, Service Bus, Blob Storage, Key Vault, Virtual Network, Managed Identity, Application Insights), Azure CLI, GitHub Actions (CI/CD, OIDC), Docker, Docker Compose
- **Linux / networking:** bash (automation; HPC cluster, PBS, conda), network administration (AP, DHCP, DNS, NAT, SSH), Cisco CCNA certificate
- **Databases:** Azure SQL, SQLite (Entity Framework Core), MongoDB
- **Security:** defense-in-depth, threat modeling, Managed Identity + Key Vault, GDPR (privacy by design)
- **Python / AI / data:** NumPy, SciPy, pandas, matplotlib, Jupyter, scikit-learn, LLM integration into applications · Psi4 (quantum chemistry)
- **Testing / processes:** TDD, integration testing (xUnit, pytest, Jest), Git / GitHub flow, code review, Scrum
- **Embedded / IoT:** Raspberry Pi (Pi 4/5, Pico), MicroPython, GPIO, sensors (IMU, encoders), I2C, serial communication

WORK EXPERIENCE

DrukMruk — Founder & Lead Developer

Sole proprietorship · 02.2026 – present · [drukruk.pl](#)

- **From zero to production in 3 months, solo** — a commercial, distributed self-service printing system for the academic community: a web app (React 19 / TypeScript), a cloud backend (ASP.NET Core 8, Azure), and a network of physical IoT kiosks (Raspberry Pi).
- Passed a **month-long security audit by a public institution (UCI NCU)** — over several rounds with the university's security team, I defended the architecture and hardened the system (network isolation of the tiers, data protection, centralized logging and incident alerting); a positive assessment was a precondition for deployment.
- Built and maintain the entire cloud backend with a **fault-tolerant payment integration (Autopay)** — cryptographically verified webhooks, automatic refunds, and guaranteed state consistency between payment and print; critical flows covered by tests (TDD); CI/CD in GitHub Actions with OIDC federation.
- Designed autonomous printing kiosks that run unattended in public spaces — Raspberry Pi 5 with a touchscreen and a local agent (FastAPI), a dedicated LTE connection, full isolation from the university network, and layered hardening (defense-in-depth) against the real-world threat model of a public location.
- Won **Copernicus Startup Stars 2026** and secured official university approval to deploy the kiosks — despite initial refusals, I carried the project through negotiations with NCU authorities, technical documentation, and the full formal / legal path.

Nicolaus Copernicus University in Toruń — Scientific programmer (quantum chemistry)

Contract · 02.2024 – 03.2026

- Designed and wrote a **Python computational library** used by the entire research group — a modular architecture that speeds up calculations relative to methods of the same accuracy class and is easy to extend; being prepared for open-source publication.
- Automated the computational cycle on the cluster, cutting a batch of calculations **from ~a day to an hour** — custom scripts for queuing and launching jobs and for resource monitoring; configured users' computing environments (Linux, conda, PBS).
- Verified the method's correctness against the scientific literature and reference data — numerical analysis (NumPy), visualizations, and comparisons for an upcoming publication.

PROJECTS

Ventilator simulator — Team leader

2026 · [github.com/yanlukashevich/Respirator-simulator](#)

- Led a 4-person team to **1st place in the NCU Institute of Physics competition** and to laureate status in the university-wide competition — a distributed, real-time ventilator simulator; the project is entering commercialization as an **NCU spin-off company**.
- Translated medical requirements into a system specification and architecture — physical patient stations (Raspberry Pi, knobs) and a trainer's panel; integrated components from different authors (Node.js / NestJS, WebSocket, React, Raspberry Pi).
- Designed a network that scales the system **from 2 to 15 stations** — built from scratch (Linux, AP / DHCP / DNS / NAT / SSH), ready to run in a training room.

EDUCATION

- **Applied Computer Science**, B.Eng. engineering studies — Nicolaus Copernicus University in Toruń, 2022–2026